Arduino UNO R4 Minima EEPROM

Learn how to access the EEPROM memory on the UNO R4 Minima.

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In this tutorial you will learn how to access the EEPROM (memory) on an **Arduino UNO R4 Minima** board. The EEPROM is embedded in the UNO R4 Minima's microcontroller (RA4M1).

Goals

The goals of this tutorials are:

- Write to the EEPROM memory,
- Read from the EEPROM memory.

Hardware & Software Needed

- Arduino IDE (online or offline)
- USB-C cable
- Arduino UNO R4 Minima
- UNO R4 Board Package

Help

EEPROM

(EEPROM) is a memory that can be used to store data that can be retrieved after power loss. This memory can be effective to use during run-time to log data that can be used to re-initialize whenever a system comes back online.

The Arduino Uno R4 Minima has 8 kB of EEPROM.

When writing to the EEPROM memory, we specify two parameters: the **address** and **value**. Each byte can hold a value between 0-255.

```
EEPROM.write(0,15); //w
```

We are writing a value of `15` to the first byte available in the memory, `0`.

To read the value of this memory, we simply use:

```
EEPROM.read(0); //reads
```

There are several more methods available when working with EEPROM, and you can read more about this in the A Guide to EEPROM.

Please note: EEPROM is a type of memory with limited amount of write cycles. Be cautious when writing to this memory as you may significantly reduce the lifespan of this memory.

EEPROM Write

A minimal example on how to **write** to the EEPROM can be found below:

```
#include <EEPROM.h>

int addr = 0;
byte value = 100;

void setup() {
  EEPROM.write(addr, v
}
void loop(){
}
```

EEPROM Read

A minimal example on how to **read** from the EEPROM can be found below:

```
#include <EEPROM.h>

int addr = 0;
byte value;

void setup() {
  Serial.begin(9600);
  value = EEPROM.read(
  while (!Serial) {

  }

  Serial.print("Addres
  Serial.println(value
}

void loop() {
}
```

Summary

In this tutorial, you've learned how to access the EEPROM on the UNO R4 Minima board. To learn more about EEPROM, visit **A Guide to EEPROM**.

Was this article helpful?